Digital Darwinism in academic libraries: survival, transformation and evolutionary paths

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Abstract

Purpose – This study aims to examine Digital Darwinism within academic libraries, focusing on institutional adaptation to technological evolution and changing user expectations. It seeks to identify disruptive forces, analyze strategic responses and develop the Academic Library Evolutionary Advancement (ALEA) framework to guide libraries in proactively managing digital transformation while upholding their fundamental values.

Design/Methodology/Approach – The paper utilizes a literature review and an analysis of current practices to examine Digital Darwinism in academic libraries. Adopting a conceptual approach, it synthesizes insights from scholarly materials, industry reports and digital resources to understand disruptive forces and adaptive strategies.

Findings – Digital Darwinism compels profound transformation in academic libraries, requiring integrated strategies. The central contribution is the four-pillar ALEA framework. This conceptual model provides a structured approach to guide libraries through digital evolution via adaptive resources, personalized experiences, empowered staff and sustainable funding.

Originality/Value – This paper's originality lies in the ALEA framework, a comprehensive, integrated model addressing Digital Darwinism, specifically within academic libraries. It moves beyond standard digitization approaches by synthesizing technology, user behavior and librarian competencies. It offers significant value by providing a theoretical basis and a practical, adaptable guide for library decision-makers to proactively manage transformation, maintain core values and ensure future relevance.

Keywords Information technology, Library users, Librarians, Individual behavior, Academic libraries, Digital Darwinism

Paper type Literature review

Introduction

The concept of Digital Darwinism applies Charles Darwin's principle of "survival of the fittest"[1] to the digital economy, suggesting that only institutions capable of adapting to rapid technological changes will thrive. First introduced by Schwartz (1999) in 1999 and later popularized by Solis (2011) in 2011, the term describes a phenomenon where technological and societal evolution outpaces an organization's ability to adapt. The digital revolution has fundamentally transformed how people access and engage with information, driving significant shifts in user expectations and behaviors. Digital platforms and technologies have redefined interactions and discourse across personal, professional and public domains. In this dynamic environment, organizations must strategically reassess their core functions and approaches to service delivery to remain relevant and impactful. Academic libraries, as institutions deeply influenced by technological progress, exemplify the need for such adaptability. Systematically integrating new technologies into their operations, these libraries face profound changes in user needs, expectations and information-seeking behaviors. To sustain their relevance and effectiveness, libraries must continually evaluate and refine their roles, service offerings and operational strategies in alignment with the demands of the digital age.

Academic libraries have traditionally been guardians of knowledge, providing information and supporting research, teaching and education (Klain Gabbay and Shoham, 2019). Over the years, they have also become integral to the cultural system, offering access to cultural resources, promoting engagement in cultural activities and fostering the development of

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information literacy (Reed and Johnson, 2023). However, the emergence of digital platforms, open-access resources and AI-driven applications has created an ongoing need for innovation in library services. As users increasingly turn to alternative information sources (Mirsultanova, 2024), libraries must evolve beyond their conventional roles. This evolution demands a strategic shift toward digital resources and user-centered approaches to maintain their relevance in today's information ecosystem.

At the same time, it is important to acknowledge that the imperative to adapt to technological change is not a neutral process. Rather, it is embedded in systemic economic and social dynamics that can produce exclusionary effects. Zuboff (2019) argues that digital capitalism does not merely reward innovation but also systematically reorganizes economic and social conditions, creating pressures that can marginalize organizations lacking sufficient technological capacity. This dynamic resembles what Arthur (1989) described as path dependency, where early adopters of digital infrastructures consolidate advantages that become increasingly difficult to challenge. Wajcman (2015) also highlights that the acceleration of digital change is shaped by power asymmetries, as those who design and control technological platforms define the standards of competitiveness. Consequently, institutions unable to keep pace with rapid innovation are often labeled obsolete and are structurally disadvantaged, regardless of their social value. In this sense, digital capitalism transforms competition into a mechanism that excludes and subordinates those unable to adapt, producing what Zuboff (2019) describes as systemic pressure toward technological superiority.

This paper examines the concept of Digital Darwinism within academic libraries, analyzing the evolutionary pressures that drive institutional change and the strategic responses essential for organizational adaptation. The research explores the transformation of traditional library functions, the integration of emerging technologies and the development of collaborative environments, while addressing the fundamental challenge of maintaining core statutory values amid digital transformation and market competition. Through a comprehensive literature review, this paper investigates the relationships between Digital Darwinism and academic libraries, with particular focus on adaptation strategies in a progressively technology-driven environment. The methodological approach involved analyzing scientific materials, industry reports and digital resources to understand the historical context, theoretical frameworks and practical implications of Digital Darwinism in academic libraries. This critical examination revealed the multifaceted nature of digital transformation, including institutional challenges, strategic opportunities and adaptive mechanisms. These insights formed the theoretical basis for our proposed framework, designed to help libraries not only survive but thrive in the digital age. This framework emphasizes a holistic approach, integrating technology, user experience, staff development and strategic partnerships.

Literature review

The concept of Digital Darwinism

Digital Darwinism, a transformative concept in business theory, is defined as "the evolution of consumer behavior when society and technology evolve faster than some companies' ability to adapt" (Solis, 2011). This concept draws parallels to Charles Darwin's theory of natural selection, suggesting that in the business world, only those companies that can effectively adapt to the digital transformation will survive and thrive, while others may face extinction. Digital Darwinism emphasizes the importance of a holistic approach to digital transformation, where organizations must integrate digital capabilities across all areas of their operations. This includes not only adopting new technologies but also rethinking business models and enhancing customer experiences to create value (Reier Forradellas and Garay Gallastegui, 2021). Digital Darwinism affects customer behaviors across all sectors, from commercial enterprises to educational institutions like universities and libraries, leading to increased brand

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sales and equity for digitally savvy organizations, emphasizing the importance of engaging with digital touchpoints (Bughin, 2014). The concept extends beyond traditional business boundaries, manifesting in fields such as culture, education, healthcare and the media and entertainment industry – each grappling with unique challenges and opportunities in the digital age.

Drivers and dynamics of Digital Darwinism

Digital Darwinism is driven by several critical factors that reshape the ways businesses and individuals interact. Chief among these is the fast advancement of technology, particularly in areas such as artificial intelligence, machine learning, blockchain and the Internet of Things. As these tools become ubiquitous, organizations are compelled to adapt to remain competitive and avoid obsolescence. This technological evolution has also revolutionized how information is accessed and consumed. The internet, functioning as an extensive repository of knowledge, enables individuals to make well-informed decisions instantaneously through their connected devices. This unprecedented access to information has intensified market competition, as consumers can effortlessly research, compare and identify optimal choices, which requires organizations to prioritize authenticity and clarity in their communications to earn trust and customer loyalty. The emergence of social media platforms and ubiquitous mobile devices has revolutionized the dynamics between content creators and consumers. These digital channels facilitate real-time, interactive engagement that traditional communication methods simply cannot replicate (Kaufmann and Manarioti, 2021).

Another critical driving force in Digital Darwinism is a transformation in consumer expectations and work paradigms. Today's consumers have evolved from passive recipients into empowered stakeholders who exert unprecedented influence in the marketplace. These digitally skilled individuals demand highly personalized experiences, expect instant gratification and require seamless omnichannel access to information and services (İpek, 2020; Jones and Kang, 2020). This shift has created a new competitive arena where traditional business models are ever more challenged by the need for real-time responsiveness and personalized engagement. The evolution of consumer behavior is reflected in several key dimensions: a growing demand for hyper-personalization across all touchpoints, the expectation of round-the-clock service availability, a preference for self-service options and digital-first interactions, a greater focus on user experience and interface design (Luther *et al.*, 2020) and a heightened emphasis on privacy and data security from both technical and legal perspectives (Widiarty and Tehupeiory, 2024). To effectively undergo this transformation, organizations must develop sophisticated capabilities across several areas. Advanced data analytics are essential for understanding and predicting consumer behavior. Automation and cloud computing, along with artificial intelligence and machine learning, enable personalization at scale, and robust customer relationship management systems supported by agile technology infrastructures are crucial for adapting to changing preferences (Pasupuleti, 2024; Sacolick, 2017; Top Digital Transformation Trends | IBM, 2024).

Intricately tied to these evolving consumer dynamics is the revolutionary shift toward remote and hybrid work models, which has significantly accelerated digital transformation initiatives across industries (Olorunyomi Stephen Joel *et al.*, 2024). This shift represents more than just technological adaptation; it encompasses fundamental changes in organizational culture, management practices, communication and collaboration methods, employee engagement and retention strategies, as well as technology infrastructure and security protocols. Organizations must prioritize continuous learning and development to equip employees with the skills needed to thrive in an increasingly digital world (Dery *et al.*, 2017). This approach ensures that the workforce remains capable of adapting to technological advancements, ultimately driving organizational success and innovation. By cultivating a culture of lifelong learning, companies can position themselves more effectively to address the challenges and seize the opportunities of digital transformation.

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Challenges and opportunities of digital transformation

Digital Darwinism presents both challenges and opportunities across various sectors. While it poses a threat to those unable to adapt, it also offers chances for innovation and redefinition of operational models. Notable failures (DigitalDefynd, 2024) and success stories (Dimitrov, 2024) serve as case studies in the corporate world, illustrating the consequences of failing to innovate and the strategies that foster resilience and growth. Companies like Blockbuster, Kodak and Toys "*R*" Us exemplify the pitfalls of complacency and resistance to change, leading to their eventual decline, while success stories such as Netflix showcase the transformative potential of embracing digital strategies and fostering a culture of innovation. The failures of prominent companies emphasize the necessity for organizations to remain vigilant and responsive to market dynamics. Blockbuster's downfall stemmed from its inability to pivot from physical rentals to streaming, while Kodak clung to traditional film, fearing the digital shift would cannibalize its core products (Satell, 2021). Similarly, Toys "*R*" Us struggled under the weight of debt and outdated business models as competitors embraced e-commerce, culminating in its bankruptcy in 2017. These cautionary tales reflect broader themes in Digital Darwinism: the dangers of neglecting consumer trends and the critical importance of adaptability in a fast-paced environment. Conversely, companies that have successfully navigated the challenges of Digital Darwinism, like Netflix or Adobe, provide valuable insights into effective strategies for innovation and customer engagement (Quixy, 2024), exemplifying successful digital transformation through technological innovation and data-driven strategies.

In the academic sector, libraries face unique challenges and opportunities that require comprehensive transformation to maintain their relevance. This transformation extends beyond digitization of collections; it necessitates a systematic restructuring of library services to meet the demands of an increasingly digital-first society. As Zharinov (2020) notes, this shift involves a deep integration of intelligent technologies within library infrastructures, enhancing user experiences through personalized content recommendations and optimized information retrieval systems, as also highlighted by Shen (2019).

This digital evolution also brings about a complex competitive landscape. Academic libraries now face three main types of competition: direct competitors, such as commercial databases; indirect competitors, like open-access repositories; and disruptive innovators, such as AI-driven research assistants and algorithmic knowledge synthesis tools. The proliferation of open-access resources – operating within various models including Green, Gold and Diamond pathways – and digital tools further intensifies this competition by providing barrier-free access to information. However, this also presents opportunities for libraries to evolve into more collaborative roles, particularly within scholarly communication networks. As Ocran and Afful-Arthur (2022) argue, academic libraries must become active partners to researchers, offering support in digital publishing and the effective utilization of digital tools.

Despite operating outside traditional commercial frameworks, academic libraries must still demonstrate their institutional value through efficient use of resources and continuous service enhancement. Their continued significance depends on evolving from traditional knowledge repositories into dynamic, innovative environments that deliver unique value propositions not available on alternative platforms. These unique offerings include expertly curated scholarly resources, authenticated information sources and collaborative learning spaces. By embracing strategic adaptations while upholding their core mission, academic libraries can position themselves as indispensable institutional pillars, fostering intellectual growth beyond the constraints of market-oriented paradigms.

The evidence presented through both cautionary tales and success stories demonstrates that Digital Darwinism is reshaping the institutional landscape across all sectors, from commercial enterprises to academic institutions. Success in this evolving environment demands three interconnected priorities: enhancing organizational adaptability, implementing comprehensive digital transformation and fostering an innovative work culture that embraces change. As technology continues to advance at an unprecedented pace,

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organizations must develop their evolutionary capabilities not merely as a strategic advantage but as a fundamental requirement for survival and growth. Whether in business or academia, the ability to adapt to digital transformation while maintaining core values and mission has become the defining factor in institutional sustainability and relevance in the digital age.

Impact of Digital Darwinism on academic libraries

As highlighted in the introductory section of this paper, academic libraries currently operate in an environment that demands continuous adaptation and self-transformation (Vollmer, 2009). Failure to adapt to these changes may result in their decline, diminished competitiveness and utility, ultimately leading to user attrition – either to alternative libraries or to services that compete with traditional library offerings. Library professionals must therefore demonstrate not only intellectual aptitude but, more critically, the capacity to meet contemporary challenges, particularly in the digital domain. This imperative is especially relevant given the year-over-year evolution in societal attitudes toward digitally available goods and services. For instance, approximately a decade ago, the following observation was still pertinent: "In the software, airline-booking, and utilities industries, consumers are more likely to be fully digital. Autos, insurance, and food have similar numbers of digital consumers in the consideration and evaluation stages, but fewer who purchase digitally. Telecommunications, banking, and appliances have relatively strong numbers of consumers considering and evaluating products and services digitally but more modest numbers making digital purchases" (Bughin, 2014, p. 2). Today, this assessment appears somewhat antiquated, given the exponential growth in digital service and product utilization. As of 2024, approximately 5.5 billion people worldwide were utilizing internet services, representing a 1.8% increase compared to the preceding year ("Internet use in 2024", 2024). Among these users, over 60% employed the internet primarily for information retrieval and maintaining communication with family and social networks. Marginally lower in prevalence were activities such as accessing current news and events, consuming video content and seeking instructional guidance for various tasks ("Reasons for using the internet worldwide Q4 2024", n.d.).

Academic libraries were established to fulfill numerous essential functions for communities within academic environments, encompassing researchers, students, practitioners from various disciplines and individuals pursuing professional development. As Shelby Foote aptly observed, "A university is just a group of buildings gathered around a library" ("Shelby Foote Quotes (Author of The Civil War, Vol. 1)", n.d.). Academic libraries have historically maintained, and continue to maintain, resources aligned with research and pedagogical requirements, providing information services related to their collections while facilitating access to materials from other institutions. These institutions conduct library instruction sessions and develop proprietary information and bibliographic tools. Furthermore, they adapt their facilities and services to accommodate users with specific needs, such as visual impairments or mobility restrictions. Academic libraries organize diverse reading-focused, promotional and exhibition events, evolving beyond mere repositories of publications into cultural and entertainment centers. Recent developments indicate a shifting focus, as evidenced by a 2021 Ex Libris™ and *Library Journal* survey, which identified the primary mission of academic libraries as fostering student engagement, particularly within remote learning environments necessitated by COVID-19 (The Pulse of the LibraryTM Report from Clarivate, 2024, p. 9). It is noteworthy that academic libraries should facilitate integrative processes that counter the social isolation trends that emerged during the COVID-19 pandemic. The widespread societal isolation has contributed to the erosion and abandonment of previously established social connections. Consequently, academic libraries, as institutions serving multiple functions – educational, scholarly and recreational – are progressively positioned to function as vital community gathering spaces. This accelerated evolution, spurred by the global shift to remote and hybrid models, has been a central theme in recent library and information science literature. Scholarly discussions, such as those featured

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in the 2022 special issue of the *New Review of Academic Librarianship* edited by Cox and Beard, have consistently examined the paradigm of accelerating digital change, underscoring the urgency for libraries to not only adopt new technologies but also fundamentally rethink their service models, organizational cultures and strategic priorities to remain relevant in a post-pandemic world (New Review of Academic Librarianship, 2022). This body of research confirms that the pressures described by Digital Darwinism are not merely theoretical but are actively shaping the contemporary operational reality of academic libraries.

Academic libraries serve – in the most general terms – to provide information and support the development of science and education at the higher level. These objectives are achievable through the implementation of various tools (The Pulse of the Library™ Report from Clarivate, 2024, p. 6) that were developed when information was scarce, its volume was still limited and access to it was restricted (Fourie, 2016). Library collections established at universities initially served narrow groups of recipients, the elite of contemporary societies. They were available – most commonly – in paper form, in specific locations and under special conditions. This resulted in academic libraries being institutions of an elite nature. They served as venues for research work and repositories of scholarly texts. Over time, they evolved into egalitarian spaces that fostered not only knowledge but also the exchange of ideas among professors and students working within them. In the 20th century, they supported research communities with book collections, periodicals and special materials, including a growing number of electronic resources toward the century's end. These developments led to academic libraries being perceived as modern institutions, receptive to technological innovations. Library spaces and equipment were subsequently adapted to accommodate the needs of database operations, resources such as e-books and audiobooks, films and games.

Electronic library catalogs were also developed during this period, accessible from outside the institutions' premises, enabling searching from any location, at any time, according to a significantly broader range of criteria than was previously possible with card catalogs (Kamińska, 2024; Ogbole and Morayo, 2017). The method of access to resources was also transformed – from hidden in storage areas to open spaces where readers could freely select their desired materials. This required librarians to reorganize resources in a way where the order/arrangement of collections corresponded to the logic and needs of users unfamiliar with the systems of organizing books and other materials on shelves. These systems were simplified, and additionally, functional and secure furniture was introduced to house the resources. Information boards with shelf and stack markings were also implemented, along with visually attractive alphabetical and subject dividers. To streamline independent external access to resources, radio frequency identification technology was introduced, which accelerated the service process and relieved librarians from this task, allowing them to focus on other activities such as providing information services to users (Bugaj, 2024; Morallo, 2018). These developments led to academic libraries being perceived as modern institutions, receptive to technological innovations. Library spaces and equipment were subsequently adapted to accommodate the needs of database operations, resources such as e-books and audiobooks, films and games.

Thus, since the end of the 20th century, academic libraries have undergone permanent changes, although such transformations had certainly occurred multiple times before. These changes were triggered by, among other factors, the revolution resulting from Johannes Gutenberg's invention of the printing press; mechanical tools were implemented from the late 1920s to expedite reader service in lending departments; and from the 1940s, photographic devices were used to archive user and book data (Jacquesson *et al.*, 1999, pp. 33–34). However, given the paper's focus, attention should be directed to contemporary developments. The first applications of computer library systems can be traced to the 1950s (batch processing), and a decade later, in 1962, work began on the Machine-Readable Cataloging (MARC) format at the Library of Congress in Washington. It was then that the first efforts toward library catalog automation were undertaken (e.g. at Florida Atlantic University in Boca Raton) (Jacquesson *et al.*, 1999, pp. 35–37). In the mid-1960s, the development of library

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networks was initiated, within which libraries were initially engaged in shared cataloging. Networks such as Online Computer Library Center (OCLC) (1965), and later Research Libraries Information Network (RLIN) (1974) and Washington Library Network (WLN) (1977), were established in the United States during this period. To meet user needs in the 1980s, thanks to work on the International Organization for Standarizaton/Open Systems Interconnection (ISO/OSI) model, various networks were interconnected, providing access to larger and larger information resources (Jacquesson *et al.*, 1999, pp. 163–180), facilitated by the implementation of the internet.

Since the 1990s, all educational institutions have experienced successive changes resulting from technological acceleration and transformations triggered by globalization processes, and consequently, the opening of service and goods markets (Fourie, 2016). Technology, initially developed and available outside libraries, led users to more readily embrace services offered in the digital market, particularly the online market. This type of service market, facilitated by the developing internet network, has forced academic libraries, among others, to compete on an unprecedented scale for readers' attention and attract them with more attractive services and tools. Beyond digital access to resources and services, competitiveness involves issues related to the restructuring of academic library organizations, including, for example, the establishment of digital media positions. Simultaneously, it became necessary to begin pursuing cost reduction in academic library operations within this new reality (Vollmer, 2009). This has enabled and continues to enable the generation of financial resources, which subsequently serve to digitize additional resources and services.

The aforementioned technological revolution is one of the drivers of change in academic libraries. The second, though related, factor is the users themselves, who have grown up in a world where they are exposed to digital tools from their earliest years, utilizing them in diverse ways – communicating with each other, organizing their daily lives, educating themselves, accessing entertainment and so forth. Significantly, the internet, including social media and various applications, is becoming the primary source of knowledge. While traditional methods of information acquisition certainly coexist, they are no longer attractive to a growing segment of academic library users (Bandyopadhyay and Boyd-Byrnes, 2016; Kumar and Gupta, 2022). The ease of unrestricted access to network resources – the ability to repeatedly search, download, transform/edit and cite them – has resulted in traditional resources, including printed materials, being less frequently utilized in the knowledge acquisition process than before. Information derived from network resources has become the first, simplest and sometimes the only source of data needed in both educational and research processes.

Academic libraries must therefore take special care to provide resources that are attractive to digitally oriented readers. The electronic collection of library resources, in turn, creates implications for adapted storage and access methods. For example, due to the lack of direct contact with library users, academic librarians must engage in developing such a system of resource access and usage instructions that ensure users do not feel lost and are motivated to utilize library resources again. These types of library survival strategies are examined more closely in the subsequent section of the paper.

Adapting academic libraries' strategies

As established in the preceding section of this paper, academic libraries have been consistently endeavoring to keep pace with technological advancements and the expanding information needs of their users. To maintain competitiveness in the market of information, educational and cultural services, these institutions must implement a comprehensive range of initiatives aimed at delivering resources and services to digital users with varying levels of information literacy. They should demonstrate three characteristics in this process: relevance, interactivity and responsibility (Vollmer, 2009). Academic libraries must now be more active in several strategic aspects – initiating collaboration, collecting digital resources, organizing access to them and creating new network-accessible services. However, to ensure that the processes of

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preventing Digital Darwinism proceed properly, appropriate research into the actual needs of library users should be conducted beforehand (Vollmer, 2009). With the cost of books continuing to rise, academic libraries are providing students with free access to educational materials (Nyemezu, 2022, p. 3).

One of the ways of coping with Digital Darwinism is through Open Educational Resources (OER), promoted by librarians specialized in this type of activity. These librarians develop access pathways to digitized materials and prepare guides to such resources and even textbooks, which allows students to significantly reduce their educational costs – without the necessity of purchasing relevant textbooks. This learning model is promoted, for example, at Florida State College Jacksonville, where students can prepare for classes using free OER materials (Esposito *et al.*, 2021). Also, students using the University of Washington Libraries' services are provided with free and licensed teaching materials ("Teaching and Learning", n.d.). However, it should be remembered that not all students are willing to use such teaching aids (Nyemezu, 2022, p. 4). Therefore, care should be taken to ensure the proper condition of traditional resources but most importantly to promote, train and prepare students to change their habits or increase their skills in using such sources in the teaching process through library training. These take on particular importance when one considers that an extremely important role of academic libraries is to support higher education. Library training enhances students'' skills in the process of searching, collecting, selecting and using the sources they find (Nowakowska and Olejnik, 2018, p. 117).

Academic libraries have expanded their technological infrastructure through the acquisition of all-in-one office devices (printers, scanners, copiers), mobile devices (laptops, tablets, e-readers), gaming consoles and related equipment. For instance, in Vilnius University Library you can use headphones, external CD-DVD drives, external memory card readers, electronic magnifiers, negative film holders, marker pens for magnetic whiteboards, USB cables for mobile devices and multifunctional audiobook media players ("Vilnius University Library", n.d.). The University of Chicago Library, on the other hand, allows the provision of equipment for videoconferences ("TechBar - Biblioteka Uniwersytetu Chicagowskiego", n.d.). This expansion necessitated strategic space allocation and resource management to ensure optimal accessibility and utilization. Library administrators faced the challenge of developing both physical and virtual environments for these resources, which initially entailed additional financial investment. No less important in this situation is to supply the library with up-to-date materials, well-equipped computers, scanners and other printing equipment (Afthanorhan *et al.*, 2019, p. 21). In their effort to maintain relevance among younger users, academic libraries have established technological laboratories, gaming studios and prototyping spaces that incorporate augmented reality technology and 3D printing capabilities (Fourie, 2016). As documented by Esposito, Campbell and Wertman, academic libraries have undergone significant transformation over the past two decades in response to societal changes. Their research highlights the implementation of innovative services, including streaming media platforms and e-books, alongside the provision of contemporary technological tools such as mobile device charging stations, tablets, Video Graphics Array adapters, calculators, digital cameras and laptops (Esposito *et al.*, 2021). This comprehensive technological integration facilitates both independent learning and collaborative project engagement. It is also indispensable to offer not only the use of mobile applications, providing convenient access to library catalogs, but also other electronic services (Yoon, 2016).

A significant portion of academic library users visit these institutions to utilize their spaces – particularly collaborative work areas that enable group project work. "Creative spaces" are emerging where readers can meet, learn together, acquire skills and exchange educational materials (Turner *et al.*, 2013). This is partly a result of the growing number of group and team assignments undertaken during studies, reflecting a change stemming from the reorganization of the teaching/learning paradigm. Academic libraries must therefore "carve out" spaces for such initiatives aimed at facilitating social activities (Turner *et al.*, 2013). This necessitates appropriate library space arrangements, with ergonomic furniture conducive to group work

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that can be easily and safely rearranged, suitable IT equipment, wireless technology, projectors, screens and lighting. Rooms for joint exercises where students can discuss freely represent a kind of connection to the ancient past of academic libraries, which, for example, in antiquity served as workplaces for scholars – sharing their knowledge, engaging in discussions and teaching. These contemporary library spaces can serve as venues for both formal and informal learning, but in each case function as physical spaces oriented toward multidirectional interaction, creativity and peer-to-peer learning. Academic libraries also provide opportunities for readers who are caring for children to use their resources. Special places are set aside for them, giving them the opportunity to use the library's reference books on the spot, and children are offered play areas, such as in Vilnius University Library ("Vilnius University Library", n.d.) and University of Warsaw Library ("Biblioteka Uniwersytecka w Warszawie", n.d.).

Academic libraries are thus becoming, like public libraries, a "third place" where young people spend time studying and engaging in recreational activities (Lotfy *et al.*, 2022). They use digital and paper resources and electronic devices, but they also relax and eat – spending time together (Lopatovska and Regalado, 2016). A rest area has been prepared, for example, at the Vilnius University Library, where you can use a microwave oven, kettle, kitchen utensils and cutlery ("Vilnius University Library", n.d.). The aforementioned change in learning methods also requires help from librarians, who serve in advisory and expert roles for users, and are ready to assist those needing specialized information (Lin *et al.*, 2015; Skórka, 2024). Users of academic libraries are predominantly young individuals who usually show a high level of skills in utilizing electronic and mobile systems (Van Wyk, 2022). However, they also exhibit highly specific expectations and demands. Consequently, library tools must be tailored to their needs, incorporating advanced data search and access systems, such as applications with streamlined biometric authorization mechanisms, including fingerprint and facial recognition (Fu, 2020; Skórka, 2024). As a result, academic librarians must already enhance their digital and communication competencies. Due to social and cultural changes, they should also be prepared for different approaches to user service than before, for example, operating on different days and hours, as implemented at the Delft University of Technology Library in the Netherlands. Due to the ethnic and religious diversity of its patrons, this institution decided to remain open year-round, without closure days (Skórka, 2024, pp. 2–3). It is important to remember such positive effects of technological transformation as the development of new specialties or rather competencies of librarians, who are becoming guides to heritage preserved in various forms and serve as navigators in the digital world (Dąbrowicz, 2024; Nemeth and Droog, 2022).

The 2024 Top Trends in Academic Libraries: A Review of the Trends and Issues ACRL report (2024) further emphasizes that academic libraries are experiencing a profound shift driven by the convergence of artificial intelligence, data-informed decision-making and heightened expectations for flexible, user-centered services. The report underscores that developing future-ready digital competencies, reimagining physical and virtual spaces and proactively engaging with ethical dimensions of technology adoption are no longer optional but essential priorities. These insights corroborate the premise that the pressures associated with Digital Darwinism are actively reshaping the strategic landscape of academic libraries.

Another survival strategy for academic libraries in times of technological change is developing an artificial intelligence implementation plan. As indicated by survey results from 1.500 libraries of various types worldwide, 60% of institutions have plans for this technology. It is primarily intended to support institutional priorities such as helping students learn (52%), improving research activities (47%) and supporting content search (45%) (The Pulse of the LibraryTM Report from Clarivate, 2024, p. 5). Artificial intelligence algorithms already enable more efficient and accurate indexing of materials stored by libraries, thereby facilitating their subsequent retrieval (Kamińska, 2024; "Transforming Libraries with Artificial Intelligence", 2024). This tool also promotes personalized user service and adaptation of library services and resources to their specific needs (Kamińska, 2024).

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Although artificial intelligence enables individual users to engage in independent learning, knowledge acquisition or even receive guidance, academic libraries can and should still focus on this advisory aspect. Librarians should assist readers in the process of information selection and curation, providing support in decision-making processes (Dąbrowicz, 2024). In addition to digital competencies, librarians should cultivate a diverse array of soft skills essential for establishing and nurturing professional relationships. These include interpersonal, communication, facilitation and coaching abilities (Dąbrowicz, 2024, p. 6). Furthermore, they must demonstrate creativity and innovation. Here, it is worth recalling that digital skills are not always considered basic; instead interpersonal skills, knowledge of professional ethics, evaluation, selection and retrieval of information, writing, teamwork and user experience are considered so (Fagan *et al.*, 2021, p. 42).

Academic librarians must simultaneously perform many roles; in addition to the standard tasks arising from library and information processes, they may also educate on information technology (although not only), collaborate with university researchers on joint research and participate in the preparation of curricula for courses related to library and information science (Fagan *et al.*, 2021, p. 40). Table 1 summarizes the key characteristics of Digital Darwinism that shape these evolving demands and expectations in the academic library environment.

The Academic Library Evolutionary Advancement Framework: guiding academic libraries through Digital Darwinism

This section introduces the ALEA Framework, a strategic model designed to guide academic libraries in navigating the challenges and opportunities presented by Digital Darwinism. The ALEA Framework, developed through a synthesis of the literature reviewed in the Literature review section, an analysis of current adaptation strategies employed by academic libraries as discussed in the Adapting academic libraries' strategies section and an understanding of emerging trends in user behavior and librarian competencies, provides a structured approach to proactively shaping the future of these institutions. It is not intended as a rigid prescription, but rather as a flexible guide that libraries can adapt to their specific contexts and resources. The following subsections will detail each of the framework's four interconnected pillars, outlining key strategies and measurable outcomes for successful implementation.

Adaptive Resource Ecosystem

The Adaptive Resource Ecosystem (ARE) pillar focuses on the library's core mission of providing access to information but reimagined for the digital age. The goal is to curate and

Table 1. Key characteristics of Digital Darwinism relevant to academic libraries

Characteristic	Description	Example impact on libraries
Technological acceleration	Rapid emergence of new tools, platforms and innovations	Continuous need to update digital services and infrastructures
Shifting user expectations	Growing demand for immediacy, personalization and mobile access	Pressure to deliver seamless, customized experiences
Competitive selection pressure	Institutions unable to adapt risk marginalization or loss of relevance	Decrease in usage, funding and institutional support
Platform-driven standards	Commercial platforms define new benchmarks for access and engagement	Necessity to adapt discovery systems, user interfaces and service expectations to match the standards set by platforms like Google, Amazon, or Netflix
Resource volatility	Unpredictable changes in funding sources, usage patterns and institutional priorities	Necessity for diversified funding and agile collection management

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provide access to a diverse, dynamic and evolving collection of resources that meet the changing needs of users. This requires a shift from a primarily print-based collection to a hybrid model that prioritizes digital resources, open access (OA) materials and data curation services. As Mirsultanova (2024) notes, users are increasingly turning to alternative information sources, making it imperative for libraries to offer compelling digital alternatives.

One key strategy within the ARE pillar is to actively diversify resource formats. Libraries should move beyond traditional print and electronic books and journals to include a wider range of digital content, such as streaming media, datasets, interactive simulations and virtual reality experiences. This diversification should be guided by user needs and preferences, as determined through surveys, focus groups and usage analytics.

Another critical component of the ARE is the prioritization of OA resources. As Esposito *et al*. (2021) point out, academic libraries are increasingly providing students with free access to educational materials to combat the rising cost of textbooks. Libraries should actively promote and support the creation, curation and dissemination of OER. This can involve partnering with faculty to develop OER textbooks, creating repositories of OA materials and advocating for OA policies at the institutional and national levels. Such assumptions necessitate proactive measures by librarians, who are tasked with identifying valuable information sources that are made available free of charge. One illustration of this can be found in the numerous sets of resources – bibliographies, repositories, digital libraries, literary text aggregators, encyclopedias, as well as both general and specialized dictionaries – currently accessible on the websites of academic libraries. By monitoring online resources, librarians identify the most valuable materials, organize them and arrange them either by subject or in alphabetical order. It should be emphasized, however, that performing this type of work only once is insufficient in the era of Digital Darwinism. Continuous review of resources is essential, including the addition of new materials and the removal of outdated ones that no longer serve as a reliable foundation for students and researchers.

It may be worthwhile to consider whether, in the context of such proactive curation of digital resources, librarians could enter into certain arrangements with creators of diverse information sources. In exchange for a form of promotion of their products, these creators could provide academic libraries with information about their work and, even if access were fee-based, offer trial versions of their databases as part of a formal agreement.

Furthermore, the ARE pillar emphasizes the importance of data curation services. As research becomes increasingly data driven, libraries have a crucial role to play in helping researchers manage, preserve and share their data. This includes providing training in data management best practices, developing data repositories and offering consulting services on data citation and metadata standards.

Metrics for success within the ARE pillar include usage statistics for different resource types, the number of OER created and adopted, researcher satisfaction with data curation services and the overall relevance of the collection as measured by user feedback. By focusing on these strategies and metrics, academic libraries can create an ARE that meets the evolving needs of learners and researchers.

Personalized user experience

The personalized user experience (PUE) pillar recognizes that today's library users expect tailored and intuitive interactions. The goal is to create personalized and engaging learning experiences that meet the individual needs and preferences of each user. This requires leveraging technology to understand user behavior, anticipate their needs and deliver customized services. As İpek (2020) notes, today's consumers demand highly personalized experiences, expecting instant gratification and seamless omnichannel access to information and services.

One key strategy for achieving PUE is the implementation of AI-powered discovery tools. These tools can analyze user search history, borrowing patterns and research interests to

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provide personalized search results and recommend relevant resources. This goes beyond simple keyword searching to offer a more intuitive and efficient discovery experience. Kamińska (2024) highlights that artificial intelligence algorithms enable more efficient and accurate indexing of materials stored by libraries, thereby facilitating their subsequent retrieval.

Customized learning experiences and interactive engagement strategies are key in enhancing PUE. Libraries can create adaptive platforms offering curated resources and interactive tutorials tailored to individual learning goals, incorporating game elements to boost engagement ("(PDF) The Potential for Using Gamification in Academic Libraries in Order to Increase Student Engagement and Achievement", 2024, p. 24). The incorporation of gamification elements – including achievement recognition systems, progression mechanics and competitive frameworks – fosters deeper connections with librarians and provides additional benefits like printer access, book vouchers or equipment. To better align with user preferences, academic libraries should place an even greater emphasis than before on interaction with users and on gathering information about their needs. Periodic, potentially cyclical surveys can serve this purpose, as can discussions – conducted both in person and via online chat, such as those at Universitets Biblioteket in Lund. A substantial amount of information regarding user preferences can also be obtained during the various library training sessions that libraries offer throughout the academic year. These sessions encompass content for new patrons as well as for individuals already familiar with the library but requiring skills related to tools they have not previously used. An example of such offerings is provided by Universitets Biblioteket in Oslo. Students, doctoral candidates and staff may participate in training for beginning readers, writing and citation, literature searches, source evaluation and selection, research data, artificial intelligence, open science, data visualization, coding and more. The broad scope of these courses ensures that users with varying levels of experience and diverse needs can find training that supports their learning and research. Therefore, librarians should pay particular attention, in their day-to-day interactions with students and researchers, to gathering information that can then be utilized to strategically tailor library activities.

Mobile-first design is also crucial for PUE. With the increasing prevalence of mobile devices, libraries must ensure that all their services and resources are optimized for mobile access. This includes developing mobile apps that provide convenient access to library catalogs, digital resources and personalized recommendations. Yoon (2016) found that user acceptance of mobile library applications in academic libraries is high when the technology acceptance model is applied.

Metrics for success within the PUE pillar include user engagement with personalized services (e.g. click-through rates on recommended resources, usage of personalized search filters), user satisfaction with learning experiences (e.g. Net Promoter Score, qualitative feedback from user interviews), mobile app usage (e.g. number of downloads, frequency of use) and completion rates for learning paths. By focusing on these strategies and metrics, academic libraries can create a PUE that fosters deeper engagement and supports individual learning success.

Empowered librarian ecosystem

The empowered librarian ecosystem (ELE) pillar recognizes that librarians are the key to successful adaptation in the digital age. The goal is to cultivate a workforce that is not only skilled in traditional librarianship but also possesses the digital fluency, adaptability and innovative mindset necessary to thrive in a rapidly changing environment. This requires investing in ongoing professional development, fostering a culture of experimentation and empowering librarians to take on new roles and responsibilities. As Dąbrowicz (2024) and Nemeth and Droog (2022) suggest, librarians are becoming guides to heritage preserved in various forms and serve as navigators in the digital world.

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To foster ELE, academic libraries should prioritize continuous learning and development for their staff. This includes providing training in emerging technologies like artificial intelligence, data analytics and virtual reality, as well as workshops on user experience design, digital marketing and project management. Furthermore, libraries ought to encourage professional growth through advanced degrees and specialized certifications. Equally important is cultivating a culture of experimentation and innovation. Librarians should be empowered to try new approaches, test new technologies and share their findings with colleagues. To facilitate this, libraries can establish internal grant programs, innovation labs and communities of practice.

A key aspect of the ELE pillar is transformation of librarian roles and responsibilities. This may involve creating new positions, such as data librarians, digital scholarship librarians or user experience librarians, or it may involve expanding the responsibilities of existing positions to include tasks such as data curation, digital content creation or online learning design.

Metrics for success within the ELE pillar include librarian participation in professional development activities (e.g. number of training hours completed, attendance at conferences), the number of innovative projects launched by librarians (e.g. new services developed, new technologies implemented) and librarian satisfaction with their professional growth opportunities (e.g. measured through employee surveys). By focusing on these strategies and metrics, academic libraries can create an ELE that supports the ongoing development and success of their staff.

Sustainable funding model

The sustainable funding model (SFM) pillar addresses the critical challenge of ensuring the long-term financial viability of academic libraries in an era of increasing costs and evolving funding priorities. The goal is to develop diversified and resilient funding streams that support the library's mission and strategic goals. This requires a shift from relying solely on traditional funding sources, such as institutional allocations, to exploring new revenue-generating opportunities and demonstrating the library's value to stakeholders. As Vollmer (2009) notes, cost reduction in academic library operations is essential for generating financial resources to digitize additional resources and services.

One key strategy within the SFM pillar is to actively pursue grant funding. Libraries should identify and apply for grants from government agencies, foundations and private organizations that support library initiatives, such as digitization projects, technology upgrades and community outreach programs. Grant writing workshops and dedicated grant officers can significantly increase the library's success rate in securing external funding.

Another important aspect of the SFM is to explore revenue-generating opportunities. This may involve offering fee-based services, such as research consulting, data analysis or training workshops, to external clients. Libraries can also generate revenue by renting out library spaces for events or conferences.

Furthermore, the SFM pillar emphasizes the importance of demonstrating the library's value to stakeholders. This involves collecting and analyzing data on library usage, user satisfaction and the impact of library services on student success and research productivity. These data can be used to advocate for increased funding from institutional administrators and other stakeholders.

Metrics for success within the SFM pillar include the amount of grant funding secured (e.g. number of successful grant applications), revenue generated from fee-based services (e.g. total revenue, number of clients served) and stakeholder perceptions of the library's value (e.g. measured through surveys, interviews and focus groups). By focusing on these strategies and metrics, academic libraries can develop a SFM that ensures their long-term financial viability and supports their ability to adapt and thrive in the digital age. Collectively, these pillars define the essential components of the ALEA Framework as a comprehensive strategy for navigating

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digital transformation. While ALEA provides a holistic model for institutional adaptation, complementary approaches offer more targeted guidance on specific aspects of technological change. For example, Cox (2024) proposes a nine-step framework focused on artificial intelligence adoption, which aligns with ALEA in emphasizing capability development, strategic alignment and continuous evaluation. Integrating insights from such specialized models can further strengthen libraries' capacity to respond proactively to emerging challenges.

Importantly, the ALEA Framework is not merely inspired by Digital Darwinism in a rhetorical sense but is structurally grounded in its core dynamics. Each pillar of ALEA directly corresponds to a dimension of evolutionary pressure that Digital Darwinism imposes on academic libraries. The ARE addresses the continuous erosion of relevance that occurs when collections fail to keep pace with users' expectations for immediacy, diversity and digital access. The PUE responds to the growing demand for hyper-personalization and seamless interaction, as expectations are increasingly shaped by commercial platforms that set new standards of engagement. The ELE counters the risk of skill obsolescence among library professionals, who must adapt rapidly to emergent technologies and evolving user behaviors. Finally, the SFM confronts the fiscal volatility inherent in digital transformation, where traditional funding streams often decline as technological investments escalate. Together, these pillars operationalize Digital Darwinism as a theoretical framework, translating its insights into actionable strategies that enable academic libraries to adapt intentionally rather than reactively. In this sense, ALEA is not simply a collection of best practices but a coherent, evidence-informed approach designed to mitigate the systemic pressures that Digital Darwinism exerts on institutions committed to sustaining their scholarly, social and cultural missions.

While analogies with corporate cases such as Blockbuster, Netflix or Amazon are frequently used to illustrate the mechanisms of technological disruption and competitive selection, it is important to acknowledge that academic and public libraries differ fundamentally in mission, governance and performance criteria. Nevertheless, recent studies confirm that libraries also face pressures akin to those experienced by commercial organizations. Shim and Jhaver (2023) demonstrate how public libraries increasingly rely on partnerships with digital content distributors to meet evolving user expectations, resulting in a hybrid service model that resembles commercial platform strategies. Similarly, Chigwada (2024) reports that academic libraries in the Global South are compelled to accelerate technological adoption and reorganize their operations to maintain relevance in post-pandemic environments.

Additional research has shown that public-sector institutions, including museums and archives, experience forms of "platformization" and market-like competition as they integrate commercial digital infrastructures to deliver content and engage audiences (Poell *et al.*, 2021). Libraries, while not profit driven, must demonstrate value, attract sustained funding and remain competitive within an ecosystem shaped by users accustomed to seamless, on-demand services. In this sense, Digital Darwinism offers a conceptual framework that helps to explain not only how commercial enterprises fail or adapt but also how public institutions are challenged to reconcile their core missions with the relentless pace of technological and social change. The alignment between the ALEA pillars and the core challenges of Digital Darwinism is further illustrated in Table 2.

By integrating the principles of the ARE, PUE, ELE and SFM, academic libraries can strategically position themselves to not only survive but thrive amid fast and continuous digital transformation. However, realizing this vision requires a keen awareness of the specific threats that Digital Darwinism presents, which will be explored in the following section. These threats require a strategic and proactive approach to adaptation, making the ALEA Framework an essential tool for academic libraries looking to succeed in an increasingly competitive information landscape.

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Table 2. Mapping ALEA pillars to Digital Darwinism challenges

ALEA pillar	Digital Darwinism challenge addressed
Adaptive Resource Ecosystem	Continuous erosion of relevance as user expectations for immediacy, diversity and digital access accelerate
Personalized User Experience	Rising demand for hyper-personalization and seamless interaction shaped by commercial platform standards
Empowered Librarian Ecosystem	Risk of skill obsolescence among library professionals due to rapid technological and social evolution
Sustainable Funding Model	Fiscal volatility caused by unpredictable changes in funding sources, institutional priorities and the escalating costs of innovation

Source(s): Authors' own work

Threats of Digital Darwinism

In the era of Digital Darwinism, academic libraries face significant evolutionary pressure to adapt or risk becoming obsolete. While the ALEA Framework, as outlined in the previous section, provides a strategic roadmap for navigating these challenges, understanding the specific threats posed by Digital Darwinism is crucial for effective implementation. The challenge isn't simply to digitize services but to fundamentally transform into competitive institutions that can effectively capture and maintain users' attention in a rapidly evolving digital landscape. Libraries must carefully balance their digital presence while being mindful of users' "digital fatigue", ensuring that their messaging remains impactful without contributing to information overflow (Bughin, 2014, p. 1).

In their digital activities, academic libraries must also consider another risk factor, namely the decreasing level of contact with readers – whether more or less digitally experienced – who perform many library operations independently, outside the library building. This way of using library services is satisfactory for many users, who value the independence of searching and using resources (Afthanorhan *et al.*, 2019, p. 21). This previously mentioned lack of direct contact between librarians and library service users requires additional efforts aimed at making the virtual library space easier and more pleasant for the latter to use.

The costs associated with technological transformation must not be overlooked in the context of academic library development. Certain modifications within academic libraries necessitate substantial financial investments allocated toward renovations, spatial reconfigurations, procurement of computing equipment, software acquisitions and other digital infrastructure. Access and licensing arrangements for commercial information resources – varying in financial impact from moderate to significant – represent another substantial expenditure category. It is essential to acknowledge that only a portion of available resources exists within OA frameworks, including scholarly journals, e-book platforms, databases, digital libraries (which have experienced continuous development since the 1970s with the pioneering Gutenberg Project) and institutional repositories. This category of resources, when integrated into library collections and catalogs, effectively reduces expenditures allocated toward the acquisition of valuable scientific information sources. Nevertheless, academic libraries, while striving to maintain resource diversity, should not restrict their collection development practices exclusively to OA materials.

Another significant challenge for academic libraries emerges from within: librarians themselves are increasingly cognizant of the necessity to undertake appropriate measures and adapt, alongside their institutions, to ongoing changes. The continued relevance of these institutions to students' and researchers' academic pursuits fundamentally depends on the librarians' capacity to evolve and innovate (Esposito *et al.*, 2021). Therefore, librarians' digital skills should be crucial for the changes occurring in academic libraries. Librarians must – if they haven't already adequately mastered digital tools – undertake appropriate training in this direction to prevent an educational gap and provide library users with more effective and

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efficient research services (Bolasco, 2023). According to the Pulse of the Library 2024 report, 52% of library employees recognize the need for training and improving their own skills in areas such as the use of artificial intelligence in professional work. These same employees, especially librarians from the United States, also express concerns about the impact of artificial intelligence on jobs (The Pulse of the LibraryTM Report from Clarivate, 2024, p. 11). Its application in daily work and user service will generate numerous challenges – beyond workshops or courses, which themselves require financial resources, additionally some librarians will need to retrain to undertake new forms of user service. Library workers may on one hand experience assistance from artificial intelligence in routine, mechanical and repetitive tasks, but on the other hand, such a form of institutional organization will consequently lead to reduced employment – which raises the greatest concern. It should not be forgotten that stress in librarians is caused not only by AI but in general by the changes taking place in the professional environment, the constant pressure to improve skills and the changing role in the workplace. The way to cope with these problems is, at the very least, to be ready to take joint actions with other people and institutions, that is, to cooperate at all levels of professional work (Aslam, 2021).

Conclusions

One must agree with the opinion of Anita Laamanen (Team Leader of eLibrary and Tools, VTT, Finland): "Over the next five years, research librarians will leverage technologies including AI to improve access to information while maintaining a strong focus on user support, ethical practices and community engagement" (*The Pulse of the Library™ Report from Clarivate*, 2024, p. 10). This statement effectively captures the balanced approach needed in modern academic libraries, where technological advancement must be harmonized with core library values and human-centered services. However, to proactively navigate this evolving landscape and ensure long-term sustainability, academic libraries should consider adopting a comprehensive framework such as the ALEA Framework.

Through the introduction of digital resources and digital services, users can experience academic libraries differently. Digitization and digitalization have radically changed the relationships between readers and institutions and their staff. Some clients have abandoned direct contact, preferring to use digital resources and services. This has required and continues to require libraries to adapt and transform – from a "stationary" library to a "mobile" one, accessible at any time and place with at least portions of their services and collections. Digital communication, digital collections, digital collaboration with library agencies and beyond, digital presence on social media platforms and institutional websites, which must include interactive notifications, access to valuable materials for viewing and downloading – these are indispensable components of a contemporary academic library. These elements align directly with the ARE and PUE pillars of the ALEA Framework, emphasizing the need for accessible and user-centric digital resources.

Academic libraries must strive to integrate their various collections and network services, but they should not forget about service users who prefer other methods of contact and library resources. This underscores the importance of the ELE pillar, where librarians play a crucial role in bridging the gap between traditional and digital services, providing guidance and support to all users.

Processes to prevent Digital Darwinism will not be effective without reader participation. Appropriate communication with them and activation to maintain contact is necessary. This, in turn, will be possible through a good understanding of their needs and expectations toward libraries. Librarians should carefully read messages left by users on social media and other communication tools, respond to them promptly and continuously monitor them. It is necessary to search for new, more sophisticated forms of anticipating and responding to readers' needs. Libraries, like other institutions, should follow the principle that "Brands today must go beyond simply broadcasting their message; they must beckon the consumer into a

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conversation" (Vollmer, 2009). This continuous feedback loop directly supports the iterative improvement of services within the ALEA Framework.

As academic librarian Małgorzata Dąbrowicz stated: "they will survive as digital libraries, virtual libraries, and as social spaces conducive to creating and maintaining interpersonal relationships" (2024, p. 7). This statement implies further changes, namely that academic libraries should be not only very modern but also oriented toward readers who need quiet, peace and relaxation – hence the necessity to organize quiet zones in institutions that prevent excessive stimulation and information overload. One solution to this problem is, for example, InnoPod® mobile acoustic pods that provide opportunities for individual work in places where numerous readers are present, such as in open-access resource areas. Through their use, one can work conveniently – in concentration, with easy access to useful sources, using personal or library equipment such as tablets or laptops ("CINIBA–Strona główna", n.d.). Another library practitioner has confirmed the earlier stated assumptions by writing: "the role of the library in the university will evolve toward content management, i.e. collecting, processing, and sharing data, information, documents, reports, etc. An opportunity worth taking advantage of today is for the library to take over activities in the area of research data sharing, open science movement, as well as collecting the publication output of university employees and related advisory services" (Skórka, 2024, p. 2).

Academic libraries must therefore keep pace with digital transformations, offering more virtualized resources and services while remaining attentive to the evolving and diverse needs of both academic and student communities. Without proactively adapting to these requirements and extending their reach to broader social circles, academic libraries may face significant challenges to their sustainability. By strategically aligning their efforts with a comprehensive framework like the ALEA, focusing on resource adaptation, user experience, librarian empowerment and sustainable funding, well-managed institutions that invest appropriate financial and organizational resources in development will not only survive, but thrive as attractive venues where users can discover engaging literature, access periodicals, benefit from professional librarian expertise, utilize electronic equipment and engage in collaborative work while maintaining social connections. Consequently, academic libraries will remain essential and attractive destinations within campus environments and university cities, actively shaping the future of information access and knowledge creation.

The ALEA Framework presented in this paper may also serve as a starting point for further discussion about the essential missions of contemporary academic libraries that are engaged in open science practices, facilitating OA to publications, research data and educational resources and supporting effective knowledge management (Liu and Liu, 2023). Libraries play a critical role in promoting civic education, fostering social inclusion and counteracting digital exclusion (Witt, 2024). Particularly important for academic libraries is the development of platforms and services that complement conventional methods of acquiring knowledge and skills, a priority that has been emphasized with increasing clarity in recent years (Santos-Hermosa, 2024).

ALEA likewise contributes to the ongoing scholarly debate about librarian literacy and professional competencies. It is essential to recognize the challenges faced by library staff, who are expected to create "third places" that shape the library experience (Wei *et al.*, 2024), support users and provide appropriate conditions for both focused work and relaxation. It should not be overlooked that librarians themselves also require training and professional development opportunities tailored to their needs and the demands they encounter (Jiang *et al.*, 2023).

Importantly, by offering a structured yet adaptable response to the pressures of Digital Darwinism, ALEA enables academic libraries to frame these activities not merely as isolated innovations but as part of a coherent strategy for institutional resilience and relevance. As technological and societal transformations accelerate, ALEA can therefore serve as both a catalyst for debate and a practical resource for understanding the changes to which libraries must adapt to continue fulfilling their vital scientific, social and cultural roles.

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Note

1. L. C. Meggison phrased it as: “It is not the most intellectual of the species that survives; it is not the strongest that survives; but the species that survives is the one that is able best to adapt and adjust to the changing environment in which it finds itself” (1963).

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